HUNDRED DAYS OF CODE
DAY 38

QUESTION: Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

SOLUTION:

1. Aim

To write a menu-driven C program using **separate functions** to perform the following operations on a string:

1. Count spaces, digits, and special characters.
2. Count the frequency of a given character.
3. Toggle the case of each alphabetic character.
4. Exit the program.

2. Algorithm

Main Algorithm

1. Start the program.
2. Input a string from the user.
3. Display the menu:
    - Count spaces, digits, and special characters.
    - Count frequency of a given character.
    - Toggle the case of each character.
    - Exit.
4. Accept the user's choice.
5. Use a switch statement to call the corresponding function.
6. Repeat the menu until the user selects **Exit**.

HUNDRED DAYS OF CODE
DAY 38

7. Stop.

Operation 1: Count Spaces, Digits and Special Characters

1. Initialize spaces, digits, and special to 0.
2. Traverse the string until '\0' is encountered.
3. If the character is a space, increment spaces.
4. If the character lies between '0' and '9', increment digits.
5. If the character is neither an alphabet, digit, nor space, increment special.
6. Display all three counts.

Operation 2: Frequency of a Given Character

1. Input the character to be searched.
2. Initialize count = 0.
3. Traverse the complete string.
4. Compare each character with the given character.
5. If they are equal, increment count.
6. Display the frequency.

Operation 3: Toggle Case

1. Traverse the string character by character.
2. If a character is between 'a' and 'z', convert it to uppercase.
3. If a character is between 'A' and 'Z', convert it to lowercase.
4. Leave digits, spaces, and special characters unchanged.
5. Display the toggled string.

C PROGRAM:

```
#include <stdio.h>


// Function declarations
void countCharacters(char str[]);
```

HUNDRED DAYS OF CODE
DAY 38

```
void characterFrequency(char str[]);
void toggleCase(char str[]);

int main()
{
   char str[100];
   int choice, i = 0;

   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);

   // Remove newline character added by fgets()
   while(str[i] != '\0')
   {
      if(str[i] == '\n')
      {
         str[i] = '\0';
         break;
      }
      i++;
   }

   do
   {
      printf("\n----- STRING OPERATIONS MENU -----\n");
      printf("1. Count spaces, digits and special characters\n");
```

HUNDRED DAYS OF CODE
DAY 38

```
printf("2. Count frequency of a given character\n");
printf("3. Toggle case of each character\n");
printf("4. Exit\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
   case 1:
      countCharacters(str);
      break;

   case 2:
      characterFrequency(str);
      break;

   case 3:
      toggleCase(str);
      break;

   case 4:
      printf("Program exited successfully.\n");
      break;

   default:
```

HUNDRED DAYS OF CODE
DAY 38

```
            printf("Invalid choice! Please try again.\n");
        }

    } while(choice != 4);

    return 0;
}



// Function to count spaces, digits and special characters
void countCharacters(char str[])
{
    int i = 0;
    int spaces = 0;
    int digits = 0;
    int special = 0;

    while(str[i] != '\0')
    {
        if(str[i] == ' ')
        {
            spaces++;
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
```

HUNDRED DAYS OF CODE
DAY 38

```
        }
        else if(!((str[i] >= 'A' && str[i] <= 'Z') ||
                (str[i] >= 'a' && str[i] <= 'z')))
        {
            special++;
        }

        i++;
    }

    printf("\nNumber of spaces = %d\n", spaces);
    printf("Number of digits = %d\n", digits);
    printf("Number of special characters = %d\n", special);
}



// Function to count frequency of a character
void characterFrequency(char str[])
{
    int i = 0;
    int count = 0;
    char ch;

    printf("Enter the character to search: ");
    scanf(" %c", &ch);
```

HUNDRED DAYS OF CODE
DAY 38

```
    while(str[i] != '\0')
    {
        if(str[i] == ch)
        {
            count++;
        }

        i++;
    }

    printf("\nFrequency of '%c' = %d\n", ch, count);
}



// Function to toggle case
void toggleCase(char str[])
{
    int i = 0;

    while(str[i] != '\0')
    {
        // Lowercase to uppercase
        if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - ('a' - 'A');
        }
```

HUNDRED DAYS OF CODE
DAY 38

```
    // Uppercase to lowercase
    else if(str[i] >= 'A' && str[i] <= 'Z')
    {
       str[i] = str[i] + ('a' - 'A');
    }

    i++;
  }

  printf("\nString after toggling case: %s\n", str);
}
```

TEST CASE TABLE:

Test Case	Input String	Operation	Expected Result
1	Hello 123!	Count characters	Spaces = 1, Digits = 3, Special = 1
2	C Programming	Frequency of g	Frequency = 2
3	Hello World	Frequency of l	Frequency = 3
4	Hello World	Toggle case	hELLO wORLD
5	CSE@2026	Count characters	Spaces = 0, Digits = 4, Special = 1
6	AbC123!	Toggle case	aBc123!
7	Any string	Exit	Program terminates

HUNDRED DAYS OF CODE
DAY 38

Test Case	Input String	Operation	Expected Result
8	Any string	Invalid choice	Displays invalid choice message

COMPILATION AND EXECUTION:



